



# NIGAPE

Your Skill Partner  
Online | Offline

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[niagape.official](https://www.instagram.com/niagape.official)



**Skill India**  
कौशल भारत - कुशल भारत





National Institute of GEN AI &  
Prompt Engineering

**nigape**

# Unlock the Power of Generative AI Lead the Future.



# ABOUT

## NIGAPE

NIGAPE was founded with a powerful vision to empower individuals in the digital era by mastering Generative AI & Prompt Engineering. We are committed to delivering practical, industry-focused AI education that prepares students and professionals for real-world challenges in an AI-driven economy.

At NIGAPE, we understand that the future of work is changing rapidly. Our programs are designed to provide hands-on training, live projects, and real implementation experience. We focus on building strong foundations in AI tools, prompt engineering, automation systems, and emerging technologies.

Our student community includes:

25% Students (After 12th & Graduates)

35% Working Professionals

25% Entrepreneurs & Freelancers

15% Career Switchers

NIGAPE is recognized for its structured curriculum, expert mentorship, and career-focused approach. Through practical learning and innovation-driven training, we aim to build India's next generation of AI leaders.

We don't just teach AI.

We prepare you to lead in the AI era.



**25000+ : Trusted Students**



**30+ : AI Tools**

**Mr. Gulshan Kumar**

**Founder**

NIGAPE



# “Most Promising Brand” IN ARTIFICIAL INTELLIGENCE TRAINING



## MISSION

At NIGAPE, our mission is to empower students, professionals, and entrepreneurs with future-ready skills in Generative AI and Prompt Engineering. We focus on practical, hands-on learning through live projects, real-world AI applications, automation systems, and industry-driven training. Our goal is to bridge the gap between traditional education and the rapidly evolving AI industry. Through structured programs, mentorship, and real implementation experience, we aim to create confident AI professionals who can build, innovate, and deliver real business impact in the digital economy.

## VISION

Our vision is to become India's leading institution in Generative AI & Prompt Engineering education. We strive to build a community of AI-driven professionals equipped with technical expertise, creative thinking, and problem-solving skills. By combining AI tools mastery, prompt engineering, automation, and real-world projects, we aim to shape leaders who can drive innovation, growth, and transformation in the global AI-powered economy. NIGAPE envisions a future where individuals don't just adapt to AI they lead with it.

**Mr. Gulshan Kumar**  
**Founder**  
(DizitalAdda)



# OUR PARTNERS COMPANIES

The logo for Tata Consultancy Services (TCS), featuring the letters 'tcs' in a stylized, lowercase font with a color gradient from orange to purple.The logo for CERTN (Handling Computer Security Incidents), featuring the word 'certn' in blue lowercase letters followed by a blue circular icon with a network-like pattern.The logo for Talent Worx, featuring a stylized 'TV' icon in purple and blue above the text 'Talent Worx' in a sans-serif font.

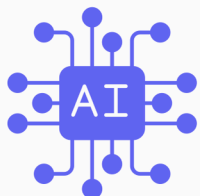
IBM Security

The logo for HCL, featuring the letters 'HCL' in a bold, blue, sans-serif font.The logo for Accenture, featuring the word 'accenture' in black lowercase letters with a small purple chevron above the 't'.The logo for APRAVA ENERGY, featuring the word 'APRAVA' in a bold, black, sans-serif font with a colorful geometric design above it, and the word 'ENERGY' in a smaller font below.

Deloitte.

The logo for Gartner, featuring the word 'Gartner' in a bold, dark blue, sans-serif font.The logo for Coforge, featuring the word 'Coforge' in a bold, sans-serif font with 'Co' in orange and 'forge' in blue.The logo for Wipro, featuring the word 'wipro' in a lowercase, sans-serif font with a colorful circular pattern of dots to its right.The logo for assystem, featuring a stylized red 'a' and 's' followed by the word 'assystem' in a lowercase, sans-serif font.The logo for VENATOR, featuring the word 'VENATOR' in a bold, blue, sans-serif font.The logo for airtel business, featuring a red circular icon with a white dot inside, followed by the word 'airtel' in red and 'business' in a smaller red font below it.The logo for LENEX, featuring the word 'LENEX' in a bold, blue, sans-serif font with the tagline 'LEARN INNOVATE EXCEL' in a smaller font below it.The logo for TECHCLOVER, featuring a stylized green and blue 't' and 'c' icon above the word 'TECHCLOVER' in a bold, sans-serif font.The logo for gridlines, featuring the word 'gridlines' in a bold, sans-serif font with 'grid' in black and 'lines' in a teal color.The logo for PHOENIX CYBER, featuring the word 'PHOENIX' in a bold, orange, sans-serif font with a stylized bird icon above the 'X', and the word 'CYBER' in a smaller, black, sans-serif font below it.The logo for birlasoft, featuring the word 'birlasoft' in a bold, sans-serif font with 'birla' in black and 'soft' in red.The logo for Netrix GLOBAL, featuring a stylized blue 'N' icon followed by the word 'Netrix' in a bold, sans-serif font and 'GLOBAL' in a smaller font below it.The logo for Palo Alto Networks, featuring a stylized orange and red icon followed by the word 'paloalto' in a bold, sans-serif font and 'NETWORKS' in a smaller font below it.The logo for nippondata, featuring the word 'nippondata' in a bold, sans-serif font with 'nippo' in orange and 'ndata' in black.The logo for MarshMcLennan, featuring a stylized blue and black icon above the text 'MarshMcLennan' in a bold, sans-serif font.The logo for Multipli, featuring a stylized purple 'm' icon followed by the word 'Multipli' in a bold, sans-serif font.

# WHY CHOOSE US ?



**AI Integrated  
Courses**



**Corporate  
Trainers**



**Practical  
Training**



**Online & Offline  
Classes**



**Corporate Based  
Environment**



**100% Placement  
Support**



**Focus Only On  
Quality Education**



**Lifetime Support**



**Live Projects**



**Updated Curriculum  
as per the Industry  
Standard**



**Case Studies**



**In House  
Internship**

# AWARDS

Indian Iconic Award By  
India's 1st Women IPS Officer Kiran Bedi



# Bharat Business Award By Ashneer Grover For India's Most Recommended Skill Based Institute



# WHO IS THIS PROGRAM FOR?



## Professionals

DizitalAdda is India's most recommended skill based institute, offering expert training in Artificial Intelligence, Data Learning and more.



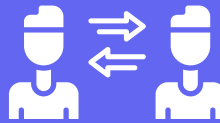
## Business Owner

DizitalAdda provides essential Artificial Intelligence skills to help business owners grow their brands online, enhance visibility, and boost sales.



## Students

A perfect opportunity for students to gain industry-relevant Artificial Intelligence skills, making them job-ready and future-proofing their careers.



## Career Switchers

Transition into a rewarding Artificial Intelligence career with practical knowledge, hands-on training, and expert mentorship at DizitalAdda.



## Freelancers

Unlock new income opportunities by mastering Artificial Intelligence skills to attract clients, grow your personal brand, and work independently.



## Start-Ups

Empower your startup with Artificial Intelligence expertise to scale your business, enhance online presence, and generate more leads effectively.

# 30+ AI TOOLS

## GET HANDS ON WITH

|                                                                                              |                                                                                                                              |                                                                                               |                                                                                                  |                                                                                                 |                                                                                                 |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|  taskade    |  fintalent.io                               |  Leonardo.Ai |  GitHub Copilot |  Blackbox AI |  you.com     |
|  Browse AI |  reclaimai                                 |  removebg   |  Katteb        |  QuillBot   |  sudo write |
|  Tactiq   |  visme                                    |  decktopus |  DURABLE      |  codeium   |  botpress  |
|  Claude   |  Gemini                                   |  Copilot   |  lumen5       |  magical   |  humanize  |
|  Cleo     |  Woebot Health                            |  Bolt      |  FLOWXO       |  tykr      |  tango!    |
|  zeta     | <b>30+ AI Tools Practical Sessions</b>  |                                                                                               |                                                                                                  |                                                                                                 |                                                                                                 |

# COURSES

## WE TRAIN

### DGAI&PE

**DIPLOMA IN GEN AI  
& PROMPT  
ENGINEERING**

### AGAI&PE

**ADVANCED  
CERTIFICATION IN GEN  
AI & PROMPT  
ENGINEERING**

**Course 1** : Diploma In Gen AI & Prompt Engineering

**Duration** : 12 Months

**Structure** :

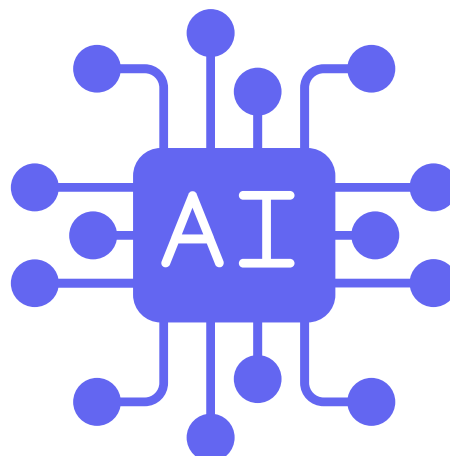
- Master the art of prompt engineering and generative AI with hands-on tools like ChatGPT, Midjourney, and DALL·E.
- The course includes AI fundamentals, advanced prompt design, real-world projects, and industry-relevant certifications.

**Course 2** : Advanced Certification In Gen AI & Prompt Engineering

**Duration** : 6 Months

**Structure** :

- Dive deep into advanced prompt strategies, AI tools, and automation workflows for real-world applications.
- Covers LLM concepts, image/video generation, chatbot building, and portfolio-based project training.



# DGAI&PE – DIPLOMA IN GEN AI & PROMPT ENGINEERING

## 12 MONTHS COURSE

### MONTH 1: AI FOUNDATIONS & SETUP

#### **Week 1: AI Foundations + Business Applications**

Basics of AI, ML, DL, GenAI ecosystem

Real-world AI use cases across industries

Types of AI tools and their business purpose

Introduction to Large Language Models

Fundamentals of Prompt Engineering

AI ethics and responsible AI practices

#### **Hands-on Implementation**

Interactive sessions with AI tools for business content generation

Exploring Perplexity.ai for industry analysis

AI dashboard exploration showing real-time applications

Generate comprehensive AI roadmap for one industry vertical

Compare different LLM outputs for same prompts

#### **Week 2: Development Environment Setup + AI Tools for Business**

Complete IDE setup (VS Code, extensions, themes)

Jupyter Notebook and Google Colab mastery

Git/GitHub workflow and version control

Professional LinkedIn and Kaggle profile creation

AI for email automation and personalization strategies

AI-driven lead generation and CRM integration

Prompt design principles for business writing

### **Hands-on Implementation**

Set up complete development environment  
Create GitHub repository with proper documentation  
Generate custom email sequences using ChatGPT  
Use Lyne.ai and Flowrite for automated follow-ups  
Create LinkedIn posts and ad headlines with Copy.ai  
Design business proposals and resumes with Notion AI  
Build cold email lists using AI-driven tools

### **Week 3: AI Content Creation Pipeline**

Presentation generation from raw concepts  
Video creation workflows from text prompts  
AI-powered course design methodologies  
Blog optimization and topic clustering techniques  
Advanced storytelling prompt frameworks

### **Hands-on Implementation**

Generate professional slide decks using Gamma  
Create video content using Pika and InVideo AI  
Develop blog summaries and social media adaptations  
Design a complete 3-module micro-course using CourseAI  
Build automated content calendar for social platforms

### **Week 4: AI Agents & No-Code Automation**

Understanding AI agents architecture and functionality  
No-code platforms deep dive (Flowise, Zapier, Bubble)  
Chatbot logic: context management, memory, conversation flow  
AI-powered web application development  
Real-world agent applications (HR, Study planning, Travel assistance)

### **Hands-on Implementation**

Build intelligent customer FAQ chatbot

Create Zapier automation workflows

Develop smart web application using Bubble

Customize Flowise agent for recommendation systems

Integration testing and deployment

## **MONTH 2: PYTHON FOR AI**

### **Week 5-6: Python Programming**

Basic Python

List , tuple , array , dictionary

Error Handling and Debugging

File I/O and Data Serialization

Python Libraries Overvie

### **Hands-on Implementation**

Build a Personal Finance Tracker

Create Data Processing Scripts

### **Week 7-8: Data Science Fundamentals**

NumPy for Numerical Computing

Pandas for Data Manipulation

Matplotlib/Seaborn for Visualization

Statistics for AI

### **Hands-on Implementation**

Exploratory Data Analysis Project

Data Cleaning and Preprocessing

## MONTH 3: MACHINE LEARNING FOUNDATIONS

### **Week 9–10: ML Algorithms & Implementation**

Supervised vs Unsupervised Learning  
Regression, Classification, Clustering  
Model Evaluation Metrics  
Cross-validation and Hyperparameter Tuning

### **Hands-on Implementation**

Implement ML algorithms from scratch  
Scikit-learn hands-on practice  
Model selection and evaluation

### **Week 11–12: Deep Learning Basics**

Neural Networks Architecture  
Backpropagation and Optimization  
Activation Functions and Loss Functions  
Introduction to TensorFlow/PyTorch

### **Hands-on Implementation**

Build Neural Networks from scratch  
Image classification with CNNs  
Text processing with basic RNNs

## MONTH 4: NATURAL LANGUAGE PROCESSING & TRANSFORMERS

### **Week 13–14: NLP Fundamentals**

Text Preprocessing and Tokenization  
Word Embeddings (Word2Vec, GloVe)  
Traditional NLP Tasks  
Introduction to Transformers

### **Hands on Implementation**

Text preprocessing pipeline  
Sentiment analysis implementation  
Named Entity Recognition

### **Week 15–16: Transformer Architecture Deep Dive**

Attention Mechanism  
Transformer Architecture (Encoder-Decoder)  
BERT, GPT Architecture  
Pre-training vs Fine-tuning

### **Hands-on Implementation**

Implement attention mechanism  
Fine-tune BERT for classification  
Text generation with GPT models  
Hugging Face Transformers library

## **MONTH 5: INTRODUCTION TO GENERATIVE AI**

### **Week 17–18: Generative Models Overview**

Types of Generative Models  
Autoregressive Models  
Variational Autoencoders (VAEs)  
Generative Adversarial Networks (GANs)  
Diffusion Models

### **Hands-on Implementation**

Implement simple GAN  
Text generation with RNNs  
Image generation experiments  
Compare different generative approaches

### **Hands on Implementation**

Text preprocessing pipeline  
Sentiment analysis implementation  
Named Entity Recognition

### **Week 15–16: Transformer Architecture Deep Dive**

Attention Mechanism  
Transformer Architecture (Encoder-Decoder)  
BERT, GPT Architecture  
Pre-training vs Fine-tuning

### **Hands-on Implementation**

Implement attention mechanism  
Fine-tune BERT for classification  
Text generation with GPT models  
Hugging Face Transformers library

## **MONTH 5: INTRODUCTION TO GENERATIVE AI**

### **Week 17–18: Generative Models Overview**

Types of Generative Models  
Autoregressive Models  
Variational Autoencoders (VAEs)  
Generative Adversarial Networks (GANs)  
Diffusion Models

### **Hands-on Implementation**

Implement simple GAN  
Text generation with RNNs  
Image generation experiments  
Compare different generative approaches

## **Week 19–20: Large Language Models (LLMs)**

Evolution of Language Models

GPT Family (GPT-1 to GPT-4)

Other LLMs (Claude, Llama, PaLM)

Model Architecture and Training

### **Hands-on Implementation**

API integration with OpenAI, Anthropic

Basic text generation tasks

Model comparison and evaluation

Understanding model limitations

## **MONTH 6: PROMPT ENGINEERING MASTERY**

### **Week 21–22: Fundamentals of Prompt Engineering**

What is Prompt Engineering?

Types of Prompts (Zero-shot, Few-shot, Chain-of-thought)

Prompt Design Principles

Common Prompt Patterns

### **Hands-on Implementation**

Design effective prompts for various tasks

A/B testing prompts

Prompt optimization techniques

Building prompt libraries

### **Week 23–24: Advanced Prompting Techniques**

Chain-of-Thought Prompting

Tree of Thought

Self-Consistency

Constitutional AI Principles

Prompt Injection and Safety

### **Hands-on Implementation**

- Implement complex reasoning chains
- Build multi-step problem solvers
- Create safe and robust prompts
- Prompt evaluation frameworks

## **MONTH 7: LANGCHAIN & AI APPLICATION DEVELOPMENT**

### **Week 25–26: LangChain Fundamentals**

- LangChain Architecture
- Chains, Agents, and Memory
- Document Loaders and Text Splitters
- Retrieval Augmented Generation (RAG)

### **Hands-on Implementation**

- Vector Databases and Embeddings
- Advanced RAG techniques
- Agent Types and Tools
- LangChain Expression Language (LCEL)

### **Week 27–28: Advanced LangChain & RAG Systems**

- Vector Databases and Embeddings
- Advanced RAG techniques
- Agent Types and Tools
- LangChain Expression Language (LCEL)

### **Hands-on Implementation**

- Build production RAG systems
- Create AI agents with tools
- Implement advanced retrieval strategies
- Performance optimization

## **MONTH 8: VECTOR DATABASES & EMBEDDINGS**

### **Week 29–30: Vector Databases Deep Dive**

- Vector Search and Similarity
- Embedding Models and Techniques
- Vector Database Options (Pinecone, Chroma, Weaviate)
- Indexing and Search Optimization

### **Hands-on Implementation**

- Set up vector databases
- Generate and store embeddings
- Implement semantic search
- Compare different embedding models

### **Week 31–32: Advanced Embedding Techniques**

- Fine-tuning Embedding Models
- Multi-modal Embeddings
- Hybrid Search Strategies
- Embedding Evaluation Metrics

### **Hands-on Implementation**

- Fine-tune embeddings for domain-specific tasks
- Build multi-modal search systems
- Implement hybrid retrieval
- Embedding quality assessment

## **MONTH 9: MULTI-MODAL AI & ADVANCED APPLICATIONS**

### **Week 33-34: Computer Vision & Image Generation**

- Convolutional Neural Networks
- Image Generation Models (DALL-E, Midjourney, Stable Diffusion)
- Vision-Language Models (CLIP, BLIP)
- Image-to-Text and Text-to-Image

### **Hands-on Implementation**

- Image classification and object detection
- Text-to-image generation
- Image captioning systems
- Visual question answering

### **Week 35-36: Multi-modal Applications**

- Vision-Language Understanding
- Audio Processing and Speech
- Video Analysis and Generation
- Multi-modal RAG Systems

### **Hands-on Implementation**

- Build multi-modal chatbots
- Implement speech-to-text systems
- Create video analysis tools
- Design multi-modal interfaces

## MONTH 10: AI APPLICATION DEPLOYMENT & MLOPS

### **Week 37–38: Model Deployment Strategies**

Deployment Architectures  
API Design and Development  
Containerization with Docker  
Cloud Deployment (Github Action / Hugging Face )

### **Hands-on Implementation**

Deploy models as REST APIs  
Create web interfaces with Streamlit  
Containerize AI applications  
Set up cloud infrastructure

### **Week 39–40: MLOps and Production Systems**

MLOps Principles and Practices  
Model Versioning and Monitoring  
A/B Testing for AI Systems  
Performance Monitoring and Logging

### **Hands-on Implementation**

Implement CI/CD for AI projects  
Set up model monitoring  
Create automated testing pipelines  
Build production-ready systems

## MONTH 11: MAJOR PROJECT DEVELOPMENT

### Week 41-42: Capstone Project Planning

Enterprise AI Assistant: Multi-functional business assistant with RAG

AI-Powered Learning Platform: Personalized education system

Creative AI Studio: Multi-modal content creation platform

Industry-Specific AI Solution: Healthcare, Finance, or E-commerce AI

### Hands-on Implementation

Project planning and architecture design

Technology stack selection

Database and infrastructure setup

Initial development and prototyping

### Week 43-44: Project Implementation

Core functionality development

Advanced feature implementation

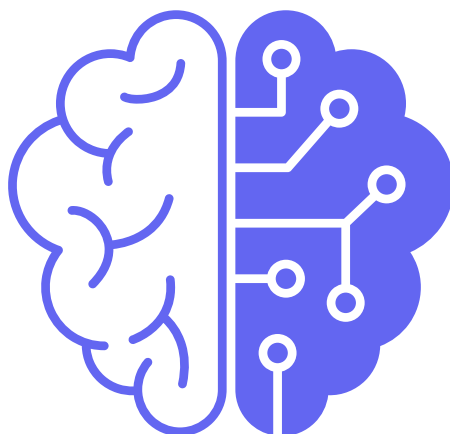
Integration testing and debugging

Performance optimization

User interface development

### Hands-on Implementation

Complete end-to-end AI application ready for portfolio



## **MONTH 12: CAREER PREPARATION & PROFESSIONAL DEVELOPMENT**

### **Week 45–46: Portfolio & Resume Building**

#### **Activities**

- GitHub profile optimization
- Project documentation and demos
- Technical blog writing
- LinkedIn profile enhancement
- AI/ML resume templates
- Achievement-focused descriptions
- Skill assessment and gap analysis
- Industry-specific resume variations

### **Week 47–48: Interview Preparation & Placement**

#### **Activities**

- AI/ML concept review
- Coding problem solving
- System design for AI applications
- Case study presentations
- Comprehensive Interview Training
- Aptitude Tests: Logical reasoning, quantitative analysis
- Coding Rounds: Python programming, algorithm implementation
- Technical Interviews: AI concepts, project discussions
- HR Rounds: Behavioral questions, salary negotiation
- Personality Development
- Communication skills enhancement
- Professional presentation skills
- Industry networking strategies
- Continuous learning mindset

# AGAI&PE – ADVANCED CERTIFICATION IN GEN AI & PROMPT ENGINEERING

## 6 MONTHS COURSE

### MONTH 1: AI FOUNDATIONS & SETUP

#### **Week 1: AI Foundations + Business Applications**

Basics of AI, ML, DL, GenAI ecosystem

Real-world AI use cases across industries

Types of AI tools and their business purpose

Introduction to Large Language Models

Fundamentals of Prompt Engineering

AI ethics and responsible AI practices

#### **Hands-on Implementation**

Interactive sessions with AI tools for business content generation

Exploring AI tools for industry analysis

AI dashboard exploration showing real-time applications

Generate comprehensive AI roadmap for one industry vertical

Compare different LLM outputs for same prompts

#### **Week 2: Development Environment Setup + AI Tools for Business**

Complete IDE setup

Jupyter Notebook and Google Colab mastery

Git/GitHub workflow and version control

Professional LinkedIn and Kaggle profile creation

AI for email automation and personalization strategies

Prompt design principles for business writing

Shopping, Stores, Google Chat, Classrooms etc.

### **Hands-on Implementation**

- Set up complete development environment
- Create GitHub repository with proper documentation
- Generate custom email sequences using ChatGPT
- Create LinkedIn posts and ad headlines with AI tools
- Design business proposals and resumes with AI assistance
- Build professional profiles on LinkedIn and Kaggle

### **Week 3: AI Content Creation Pipeline**

- Presentation generation from raw concepts
- AI-powered content creation methodologies
- Blog optimization and topic clustering techniques
- Advanced storytelling prompt frameworks
- Social media content automation

### **Hands-on Implementation**

- Generate professional slide decks using AI tools
- Create blog content and social media posts
- Develop automated content calendar
- Build comprehensive content creation workflow
- Design marketing materials using AI

### **Week 4: AI Agents & Basic Automation**

- Understanding AI agents architecture and functionality
- No-code platforms introduction
- Basic chatbot logic and conversation flow
- Simple automation workflows
- Real-world agent applications

### **Hands-on Implementation**

- Build intelligent customer FAQ chatbot
- Create basic Zapier automation workflows
- Develop simple AI-powered tools
- Integration testing and basic deployment
- Documentation and version control

## **MONTH 2: PYTHON FOR AI DEVELOPMENT**

### **Week 5-6: Python Programming Essentials**

- Python syntax and basic programming concepts
- Data types: Lists, tuples, dictionaries, sets
- Control structures: loops, conditionals
- Functions and modules
- Error handling and debugging
- File I/O operations
- Python libraries overview

### **Hands-on Implementation**

- Build a Personal Finance Tracker
- Create Data Processing Scripts
- Develop utility functions for AI projects
- Practice with real-world coding challenges
- Build reusable code modules

### **Week 7-8: Data Science with Python**

- NumPy for numerical computing
- Pandas for data manipulation and analysis
- Matplotlib/Seaborn for data visualization
- Basic statistics for AI
- Data cleaning and preprocessing
- Working with CSV, JSON, and other data formats

### **Hands-on Implementation**

Exploratory Data Analysis projects  
Data cleaning and preprocessing pipelines  
Create interactive visualizations  
Build data analysis reports  
Work with real business datasets

## **MONTH 3: MACHINE LEARNING FOUNDATIONS**

### **Week 9–10: Introduction to Machine Learning**

What is Machine Learning ?  
Types of ML: Supervised, Unsupervised, Reinforcement Learning  
Basic algorithms: Linear Regression, Decision Trees, K-Means  
Model evaluation metrics (Accuracy, Precision, Recall)  
Train-test split and cross-validation  
Introduction to Scikit-learn

### **Hands-on Implementation**

Build first ML model & Implement classification tasks  
Implementation of Email Spam Classification  
Model comparison and evaluation  
Visual model performance analysis

### **Week 11–12: Deep Learning Basics**

Introduction to Neural Networks  
Basic concepts: neurons, layers, activation functions  
Introduction to TensorFlow/Keras (basic usage)  
Simple neural network architectures  
Training and evaluation of neural networks  
Overfitting and regularization basics

### **Hands-on Implementation**

Build a simple neural network from scratch  
Image classification with basic CNN  
Text classification with simple models  
Compare traditional ML vs Deep Learning  
Visualize neural network training

## **MONTH 4: NATURAL LANGUAGE PROCESSING & LLMs**

### **Week 13–14: NLP Fundamentals**

Introduction to Natural Language Processing  
Text preprocessing: tokenization, cleaning, normalization  
Bag of Words and TF-IDF  
Sentiment analysis basics  
Named Entity Recognition (NER)  
Introduction to word embeddings

### **Hands-on Implementation**

Build text preprocessing pipelines  
Create sentiment analysis tool  
Implement basic chatbot  
Text classification projects  
Word cloud and text visualization

### **Week 15–16: Introduction to Transformers and LLMs**

What are Transformers? (High-level overview)  
Introduction to BERT and GPT models  
Understanding pre-trained models  
Hugging Face Transformers library  
Fine-tuning basics (concept level)  
API integration with popular LLMs

### **Hands-on Implementation**

- Use pre-trained models for text classification
- Integrate OpenAI API for text generation
- Build simple question-answering system
- Compare different LLM outputs
- Create text summarization tool

## **MONTH 5: PROMPT ENGINEERING & AI APPLICATIONS**

### **Week 17–18: Prompt Engineering Fundamentals**

- What is Prompt Engineering?
- Types of prompts: Zero-shot, Few-shot, Chain-of-thought
- Prompt design principles and best practices
- Common prompt patterns and templates
- Prompt optimization techniques
- A/B testing for prompts

### **Hands-on Implementation**

- Design effective prompts for various tasks
- Build prompt template library
- Create prompt optimization workflows
- Implement A/B testing for prompts
- Develop prompt evaluation metrics

### **Week 19–20: Advanced Prompting & LangChain Basics**

- Chain-of-Thought prompting
- Role-based prompting
- Constitutional AI principles
- Introduction to LangChain
- Basic chains and agents
- Document processing with LangChain

### **Hands-on Implementation**

- Implement complex reasoning chains
- Build multi-step problem solvers
- Create document Q&A systems
- Develop basic LangChain applications
- Build conversational AI agents

### **Week 21-22: RAG Systems & Vector Databases**

- Introduction to Retrieval Augmented Generation (RAG)
- Vector databases concepts (Chroma, FAISS)
- Embeddings and semantic search
- Document loaders and text splitters
- Basic RAG implementation
- Evaluation of RAG systems

### **Hands-on Implementation**

- Build basic RAG system
- Implement semantic search
- Create document chat interface
- Set up vector databases
- Develop knowledge base systems

### **Week 23-24: AI Application Development**

- Building AI-powered web applications
- Streamlit for rapid prototyping
- Gradio for ML model interfaces
- API development basics
- User interface design for AI apps
- Basic deployment strategies

## **Hands-on Implementation**

Create AI web applications with Streamlit

Build model interfaces with Gradio

Develop API endpoints for AI models

Design user-friendly AI interfaces

Deploy applications to cloud platforms

# **MONTH 6: LANGCHAIN & AI APPLICATION DEVELOPMENT**

## **Week 25–26: LangChain Fundamentals**

LangChain Architecture

Chains, Agents, and Memory

Document Loaders and Text Splitters

Retrieval Augmented Generation (RAG)

## **Hands-on Implementation**

Vector Databases and Embeddings

Advanced RAG techniques

Agent Types and Tools

LangChain Expression Language (LCEL)

## **Week 27–28: Advanced LangChain & RAG Systems**

Vector Databases and Embeddings

Advanced RAG techniques

Agent Types and Tools

LangChain Expression Language (LCEL)

# Generative AI for Professional

**16 Weeks**

## Module 1: Foundations

### **AI Business Landscape**

Introduction to Generative AI in Business  
AI vs Traditional Software: Business Impact Analysis  
Market Trends and Competitive Landscape  
Success Stories and Case Studies Analysis  
Strategic Framework Development  
AI Maturity Assessment Models  
Business Value Identification Techniques  
Stakeholder Alignment Strategies  
Hands-on Workshop  
Business Case Development Exercise  
ROI Calculation Templates and Tools

### **AI Business Landscape**

Understanding Large Language Models  
Architecture Overview: Transformers, GPT, BERT  
Capabilities and Limitations Analysis  
Model Selection Criteria for Business Use Cases  
AI Ethics and Governance  
Bias Detection and Mitigation Strategies  
Data Privacy and Security Frameworks  
Regulatory Compliance (GDPR, AI Act)  
Lab Session  
Model Comparison and Evaluation  
Setting up Development Environment

## Module 2: Strategic Implementation

### **AI Strategy & Roadmap**

Strategic Planning Framework  
Current State Assessment Methodologies  
Gap Analysis and Opportunity Identification  
Resource Planning and Budget Allocation  
Implementation Roadmap Design  
Phased Rollout Strategies  
Risk Assessment and Mitigation Plans  
Success Metrics and KPI Definition  
Strategy Workshop  
Create personalized AI adoption roadmap  
Stakeholder presentation preparation

### **Change Management & Leadership**

Organizational Transformation  
Change Management Models for AI Adoption  
Team Structure and Role Definition  
Training and Upskilling Strategies  
Executive Communication  
C-Suite Presentation Techniques  
Financial Modeling for AI Investments  
Board-Level Reporting and Governance  
Leadership Simulation  
Executive presentation practice  
Q&A handling and objection management

## **Vendor & Technology Selection**

Market Analysis and Vendor Evaluation  
AI Platform Comparison Matrix  
Build vs Buy Decision Framework  
Contract Negotiation Strategies  
Technical Due Diligence  
Security and Compliance Assessment  
Integration Complexity Analysis  
Scalability and Performance Planning  
Procurement Workshop  
RFP development for AI solutions  
Vendor evaluation scorecards

# **Module 3: Technical Implementation**

## **API Integration & Development**

API Fundamentals  
OpenAI API integration and authentication  
Rate limiting, error handling, and best practices  
Cost optimization strategies  
LangChain Framework  
Chain development for complex workflows  
Memory management and conversation handling  
Custom tool integration and function calling  
Development Lab  
Build a chatbot with memory and context  
API cost monitoring dashboard

## **Advanced AI Applications**

Retrieval-Augmented Generation (RAG)  
Document processing and vectorization  
Vector database setup and management  
Semantic search implementation  
Multi-Modal AI Integration  
Image and document analysis  
Audio processing and transcription  
Cross-modal content generation  
Advanced Lab  
Knowledge base chatbot development  
Multi-modal content analysis system

## **Enterprise Integration**

System Architecture Design  
Microservices architecture for AI applications  
Database integration patterns  
API gateway and security layers  
Workflow Automation  
Business process automation with AI  
Integration with CRM, ERP, and other systems  
Event-driven architecture patterns  
Integration Project  
Design enterprise AI architecture  
API integration with business systems

## **Performance & Monitoring**

System Performance Optimization

Caching strategies and performance tuning

Load balancing and scaling techniques

Cost optimization and resource management

Monitoring and Observability

AI system monitoring frameworks

Alert systems and incident response

Performance metrics and SLA management

Monitoring Workshop

Complete monitoring dashboard setup

Performance optimization exercise

# **Module 4: Business Applications**

## **Marketing & Content Applications**

API Fundamentals

OpenAI API integration and authentication

Rate limiting, error handling, and best practices

Cost optimization strategies

LangChain Framework

Chain development for complex workflows

Memory management and conversation handling

Custom tool integration and function calling

Development Lab

Build a chatbot with memory and context

API cost monitoring dashboard

## **Customer Service & Support**

Intelligent Chatbot Development  
Advanced conversation design patterns  
Escalation handling and human handoff  
Multi-language support implementation  
Support Process Automation  
Ticket classification and routing  
Knowledge base management  
Customer sentiment analysis  
Support System Project  
Complete customer service AI system  
Integration with help desk platforms

## **Sales & CRM Applications**

Sales Process Enhancement  
Lead scoring and qualification  
Personalized proposal generation  
Sales forecasting and pipeline analysis  
CRM Integration and Automation  
Contact management and enrichment  
Follow-up automation and scheduling  
Performance analytics and reporting  
Sales Automation Project  
End-to-end sales process automation  
CRM integration and dashboard

## **Operations & Process Optimization**

Document Processing and Analysis  
Contract analysis and extraction  
Invoice processing automation  
Compliance monitoring and reporting  
Decision Support Systems  
Predictive analytics integration  
Risk assessment automation  
Strategic recommendation engines  
Operations Project  
Process automation and optimization  
Decision support dashboard

## **Module 5: Advanced Topics & Capstone**

### **Emerging Technologies & Future Trends**

Cutting-Edge AI Technologies  
Multimodal AI and autonomous agents  
Code generation and software development AI  
Industry-specific AI applications  
Future-Proofing Strategies  
Technology roadmap planning  
Continuous learning frameworks  
Innovation management processes  
Innovation Workshop  
Future scenario planning  
Technology adoption strategies

## Capstone Project & Certification

Capstone Project Development

Individual or team-based comprehensive project

Real business problem solving

End-to-end solution implementation

Project Presentations

Professional presentation to industry panel

Peer review and feedback sessions

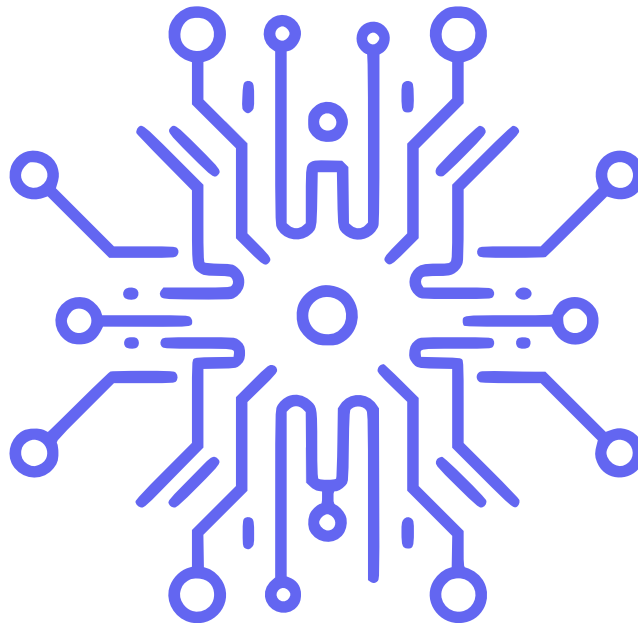
Best practices sharing

Certification & Career Planning

Final assessment and certification

Career transition planning

Alumni network and continuing education



# CHOOSE THE BEST PROGRAM

| Topics                                                 | 6 Months Course | 1 Year Course   |
|--------------------------------------------------------|-----------------|-----------------|
| AI Foundation & Setup                                  | ✓ Includes      | ✓ Includes      |
| Python for AI Development                              | ✓ Includes      | ✓ Includes      |
| Machine Learning Foundations                           | ✓ Includes      | ✓ Includes      |
| Natural Language Processing ,<br>Transformers and LLMs | ✓ Includes      | ✓ Includes      |
| Prompt Engineering & AI<br>Applications                | ✓ Includes      | ✓ Includes      |
| Generative AI Foundations                              | ✓ Includes      | ✓ Includes      |
| LangChain & AI Application<br>Development              | ✗               | ✓ Includes      |
| Vector Databases & Embeddings                          | ✗               | ✓ Includes      |
| Multi-modal AI & Advanced<br>Applications              | ✗               | ✓ Includes      |
| AI Application Deployment &<br>MLOps                   | ✗               | ✓ Includes      |
| Major Project Development                              | 1-Major Project | 2-Major Project |

# PLACEMENT SUPPORT FEATURES



**Career Guidance**



**Resume Building**



**LinkedIn Profile  
Building**



**In House Interview  
Preparations**



**Interview Sessions**



**Extensive Network,  
Growing Opportunities**



**Personal Development  
Session**



**Confidence Buildup**

# MASTER AI TOOLS FOR THE FUTURE OF WORK

In today's rapidly changing world, our program equips you with the practical skills you need to succeed. We go beyond theory, diving deep into the latest AI tools that are transforming industries. Learn to leverage this powerful technology and gain a competitive edge in the job market.

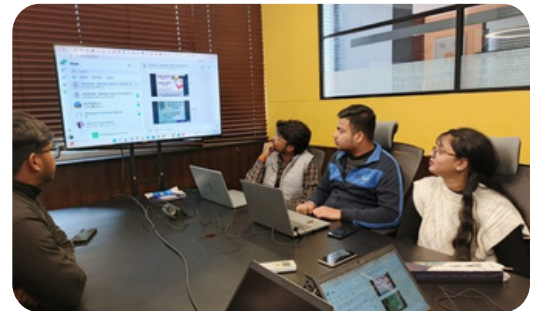
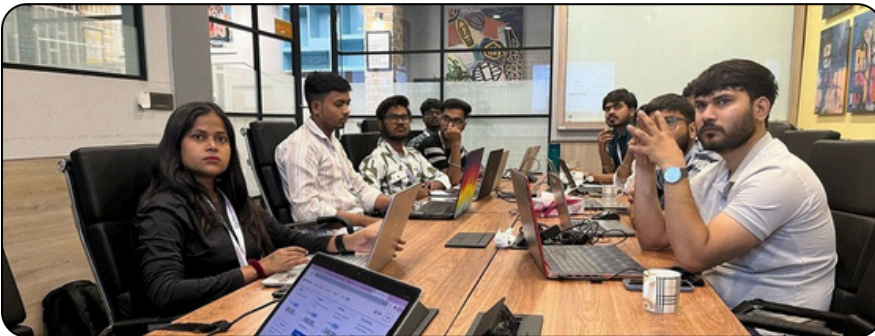
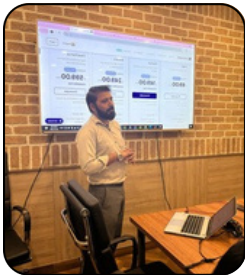
## Unleashing AI's Potential: The Power of Prompt Engineering

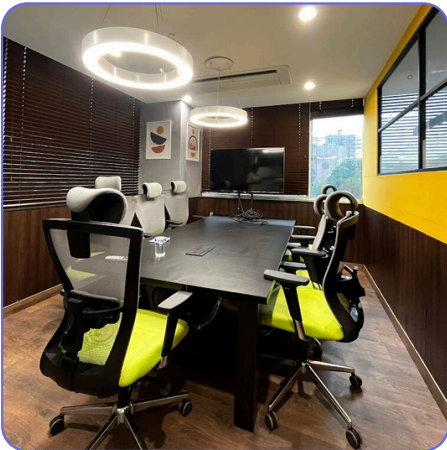
Prompt engineering empowers users to become co-pilots with AI, navigating its capabilities to achieve remarkable results. As AI continues to evolve, prompt engineering will become an essential skill, unlocking its full potential and shaping the future of human-AI collaboration. It's a game-changing feature that unlocks the true potential of AI by allowing users to fine-tune their interactions and achieve more precise, relevant outputs.

### AI DRIVEN TOOLS



# CORPORATE CLASSES





# CAREER OPPORTUNITIES

After Completing Our Generative AI Training

## INTERNSHIP/JOB

- Prompt Engineer
- AI/ML Engineer
- Deep Learning Engineer
- AI Content Creator
- AI Designer / AI Artist
- AI Trainer / Data Annotator
- AI Product Manager
- AI Security Specialist

### ENTREPRENEURSHIP

- Improved Online Presence

## FREELANCE GENERATIVE AI WORK



## CONTENT CREATION/INFLUENCER



## AFFILIATE MARKETER



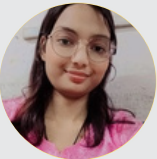
# WHAT OUR STUDENTS SAY



NIGAPE offers top-notch AI , ML , GenAI training with expert corporate trainers. Highly recommended.



Great institute with top-quality education, practical exposure, and expert trainers. Loved the experience.



Best institute for AI Quality education, real-world training, and a great corporate vibe.



Corporate-level training with advanced tools and real projects. Best place to master Artificial Intelligence.



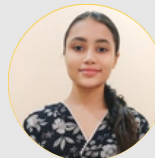
The trainers at NIGAPE are industry experts. The corporate-style learning is a game-changer.



NIGAPE bridges the gap between theory and practice. The corporate learning style is outstanding.



NIGAPE provides excellent hands-on training with live projects. Perfect for career growth.



I got hands-on experience with AI Best place to start a Artificial Intelligence career.



The corporate environment at NIGAPE makes learning Artificial Intelligence feel like real industry work.



NIGAPE provides a real corporate feel with interactive sessions and live projects. Highly recommended.



Best place to learn AI Corporate trainers provide deep insights into real-world strategies.



Perfect learning experience! Industry-focused curriculum, expert trainers, and great mentorship.



NIGAPE structured course and practical learning helped me land my first AI job.



Best decision to join NIGAPE! Quality training, corporate atmosphere, and real-world case studies.

# NOTES

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# NIGAPE

## GET IN TOUCH

# WE

We help students and professionals build real world skills in Generative AI, Prompt Engineering, AI tools, and emerging technologies through practical learning, live projects, and industry-focused training.

## Contact us:

2nd Floor, Spacetime Management Pvt Ltd Design House, behind Savitri Cinema Complex, Greater Kailash II, Chittaranjan Park, New Delhi, Delhi 110048

